

The circular-arc Śrī Yantra

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Abstract

The Śrī Yantra is the union of nine triangles, symmetric about a common axis, whose sides meet in a large number of prescribed coincidences. Chiodo showed that the concurrent models admit a straightedge and compass construction, and that the problem is equivalent to a circle–line–point problem of Apollonius. We consider the configuration obtained by replacing each of the twenty-seven sides by a circular arc, keeping every incidence. We give a straightedge and compass construction of it: twelve seed points carrying eighteen free real parameters, followed by forty-five steps, each of which is a circle through three points or the meeting of two circles. Chiodo’s concurrency conditions are satisfied identically — each one is a coincidence of nodes of the configuration rather than a numerical near-miss — so every choice of the eighteen parameters for which the construction is defined yields an exact circular-arc Śrī Yantra. The construction strengthens the classical incidences rather than merely transposing them: it places all fourteen outer corners on the circumscribed circle, where Chiodo’s condition (i) requires only the six corners of t_3 and t_7 . We compute the dimension of the family in two independent ways, by counting the parameters of the construction and by computing the rank of the incidence system, and obtain 18 both times. Two subfamilies, singled out by asking that the outer vertices of one of the two grown rings of cells be concyclic, have dimensions 12 and 11. Neither can be reached by a construction in which every step is forced by the preceding ones, and we give a certificate of this that can be checked by hand; but each is reached by a construction that departs from that pattern at exactly one point, where one number is chosen so that one coincidence holds, which is also the character of Chiodo’s construction of the straight figure. Whether that one coincidence can itself be arranged with straightedge and compass we leave open, and we locate the question precisely: the coincidence can sit at only eleven objects of the figure, and at two of them it is three compass strokes from being drawn. An interactive viewer and three GeoGebra files accompany the paper.

1 Introduction

The Śrī Yantra, or Śrī Cakra, is a diagram of Tantric Hinduism: a central point, the *bindu*; around it an interlocking figure of nine triangles; around that a ring of eight lotus petals, a ring of sixteen, three circles, and the triple square called the *bhūpura*. The nine-triangle figure is the mathematically interesting part, and it is the part treated here; Figure 1 shows one of the figures constructed below.

Drawing it accurately is a genuine problem. The nine triangles are all isosceles and share a vertical axis of symmetry; their sides are required to pass through a large number of common points, and these requirements are not independent. Traditional methods, transmitted by commentators on the *Saundaryā Laharī*, satisfy some of the conditions

*Claude Opus 5, model string `claude-opus-5`, run as an agent with file, shell and browser access at maximum reasoning effort. Work carried out in August 2026.

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and miss others, and the discrepancies are visible to the eye. The modern literature on the problem is substantial: Bolton and Macleod [1], Kulaichev [7], Fonseca [3], Tahta [12], Rao [9], Huet [6] and Pustynski [8] each treat the constraints and the accuracy with which they can be met, and Gupta [5] surveys the yantras of mathematical interest more broadly.

The question was settled by Chiodo [2], who gave a straightedge and compass construction of the concurrent models and identified the essential step as a circle–line–point problem of Apollonius. Chiodo’s paper also fixes the notation we use here, and we follow it: the nine triangles are $t_1, \dots, t_9$, numbered by the height of their bases from the top, with t_i pointing downwards exactly when $i \leq 5$; the *base point* of an isosceles triangle is the midpoint of its base; and its *legs* are the two sides exchanged by the reflection.

Why arcs

Straight sides are not the only tradition. The *Gaurīyāmala Tantra*, quoted by Gupta [5, pp. 181–182], names four architectural forms (*prastāra*) of the figure: *bhū*, the plane form, in which the whole diagram lies in a horizontal plane; *kūrma*, likened to the back of a tortoise, in which the triangular complex is drawn on a spherical surface using spherical triangles; *meru*, in which the enclosures lie in horizontal planes at different heights, after the mythical mountain; and *padma*, the lotus, of which Gupta records only that it does not seem to be popular. The present paper, like Chiodo’s, treats the *bhū* form.

The two described non-plane forms differ in kind. In *kūrma* the object remains a surface figure: it lies on a 2-manifold, with its sides geodesics of that manifold, which is the setting of the second question of Section 9. In *meru* the enclosures occupy horizontal planes at different heights, so the figure does not lie on one surface at all, and is three-dimensional in a stronger sense. Rao [9] gives a formulation of the *kūrma* form and optimises it numerically over a range of cap depths, parametrised by an altitude h running from 0 to $\pi/2$; he reports the global optimum near $h = \pi/2$, which is the limit in which the figure occupies a vanishing cap and the plane form is recovered.

The identification of *kūrma* with the sphere deserves a caveat. The name and the image are those of a tortoise’s carapace, which is domed but not spherical; Gupta passes from the one to the other within a single sentence, and Rao’s formulation is spherical throughout. If the identification is a later convenience, then the surface sought in the second question of Section 9 is not merely a mathematical generalisation of the spherical case: it may be nearer to what the classification named.

A physical instance is the Śrī Cakra carved on the crown of the śivaliṅga at the Chintamaniśvara temple, Anegundi, Koppal district, Karnataka.¹ The crown is a shallow dome rather than a hemisphere. Its innermost triangles are close to straight and the bowing grows outwards, as for a figure laid on a gently curved surface; the result is markedly less distorted than the deeper spherical optima of [9, Fig. 5].

The configuration studied below is not the *kūrma* variant — our arcs lie in the plane, not on a cap — but the two share the thought that the sides of the yantra need not be straight, and that what matters is the incidence structure rather than the straightness. For a shallow cap the two are close: a geodesic there is a short arc near the pole, its orthographic projection is a short piece of an ellipse, and a short elliptical arc is approximated by a circular one to high accuracy.

There is also a purely mathematical reason. The straight figure is rigid in a strong sense: Chiodo’s conditions leave a family of small dimension, and the concurrences are

¹Photographs of the liṅga, from above and in profile, are collected in the spherical gallery of [11].

hard to satisfy because there is little room to move. Allowing each side to bow — one extra parameter per side, the curvature — enlarges the family considerably, and the interesting question becomes what the incidence structure looks like when there is room to spare. It turns out to look very simple: the conditions can be met not approximately but *identically*, by a construction in which each required coincidence is a single node used twice.

What is proved

Let Γ denote the incidence structure of the nine-triangle figure: nine triangles, twenty-seven sides, and the sixty-nine nodes at which sides are required to meet. We prove:

- There is a straightedge and compass construction Φ taking twelve seed points, carrying 18 free real parameters, to a realisation of Γ by circular arcs (Section 4). Every step is a circle through three points, or the meeting of two circles, of a circle with the axis, or of a circle with the rim.
- Every figure in the image of Φ satisfies Chiodo’s conditions (i), (ii) and (iii) exactly, and does so combinatorially: each of the seven conditions of type (ii) and each of the twelve of type (iii) holds because two objects of the configuration share a node (Theorem 5.1).
- More holds than Chiodo’s conditions ask. The construction places fourteen corners on the circumscribed circle, against the six of condition (i); fourteen is the number of corners lying on the boundary of the figure, so no realisation can do better (Theorem 5.1(d)).
- The family has dimension 18. This is the count of free parameters in the construction; it is confirmed independently by the rank of the incidence system, which has 114 equations of rank 88 in 106 unknowns (Theorem 6.1).
- Outside the nine triangles the figure grows two rings of cells. Asking that the outer vertices of the first ring be concyclic is 6 independent conditions and leaves a subfamily of dimension 12; asking it of the second ring is 7 conditions and leaves dimension 11 (Theorem 7.2).
- Neither ring condition can be built into a construction *internal* to the figure — one in which every object drawn is a side, a node or a ring tip of it, each produced from exactly as many earlier objects as determine it. The obstruction is a set of 36 (resp. 37) objects each over-determined by the others, which the reader can verify (Proposition 7.5). Allowing one object to be over-determined — one *arranged coincidence*, where a parameter is chosen so that a further incidence holds — a construction exists for each ring, and one is the minimum (Proposition 7.6). The classical straight figure has the same count, its one coincidence being the Apollonius problem of Chiodo’s construction. Whether the arc figures’ coincidence can likewise be arranged with straightedge and compass is left open; Section 7.2 states the question exactly and reports the evidence.

Section 8 describes the three GeoGebra files that accompany the paper, one for a distinguished member of each family.

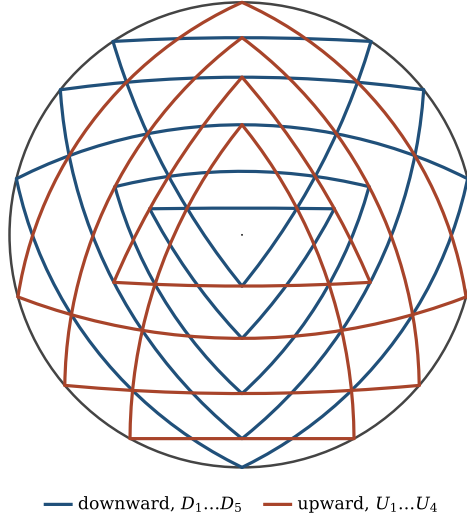


Figure 1: A circular-arc Śrī Yantra: the union of nine circular-arc triangles, five pointing down and four up, all symmetric about the vertical axis. The outer circle is the rim; the bindu is the point at the centre. This is the figure carried by the file `sri-yantra-arcs-free.ggb`.

2 The configuration

2.1 Chiodo's conditions

We recall the conditions in the form given in [2, pp. 377–378]. Let $t_1, \dots, t_9$ be the nine triangles, numbered by the height of their bases from the top, so that t_i points downwards for $i \leq 5$ and upwards for $i \geq 6$. Write $\text{apex}(t)$ for the vertex opposite the base and $\text{bp}(t)$ for the base point.

These nine are the triangles bounded by the twenty-seven full sides; the figure is their union, and every other triangle visible in the diagram is a part of one of them, arising from the way the nine cut one another. They are not nested in any regular way, and several lie inside others: at the distinguished figure of Section 8, t_5 lies wholly within each of t_2, t_3, t_4, t_7 and t_8 , and t_4 and t_6 each lie within three of the rest.

- (i) t_3 and t_7 are inscribed in one and the same circle.
- (ii) $\text{apex}(t') = \text{bp}(t'')$ for each of the seven ordered pairs

$$(t', t'') \in \{(t_8, t_1), (t_6, t_2), (t_9, t_3), (t_1, t_6), (t_5, t_7), (t_4, t_8), (t_2, t_9)\}.$$

- (iii) On each side of the axis, a leg of the downward triangle t' , a leg of the upward triangle t'' and the base of t'' have exactly one point in common, for each of the twelve triples

$$(t', t'', t''') \in \{(t_1, t_2, t_7), (t_2, t_3, t_7), (t_1, t_3, t_8), (t_1, t_4, t_6), (t_1, t_5, t_9), (t_4, t_6, t_9), \\ (t_2, t_7, t_9), (t_3, t_7, t_8), (t_3, t_8, t_9), (t_4, t_4, t_8), (t_5, t_5, t_6), (t_2, t_6, t_6)\}.$$

Following [2] we write these as (i), ((ii); t_i, t_j) and ((iii); t_i, t_j, t_k). Conditions (ii) and (iii) are 7 and 12 statements respectively; by the mirror symmetry each statement of type (iii) is one condition, not two.

2.2 Nodes and sides

It is convenient to record the combinatorics once and for all. Each triangle contributes three sides — a base and two legs — so there are 27 sides. The points at which sides are required to meet, together with the corners of the triangles, are the *nodes*; there are 69 of them. The reflection in the axis acts on all of this: it fixes 9 nodes (those on the axis) and exchanges the remaining 60 in 30 pairs; it fixes 9 sides (the bases, which are symmetric) and exchanges the remaining 18 legs in 9 pairs.

Fourteen nodes lie on the rim: the two poles, which are on the axis, and six mirror pairs. All fourteen are corners of the nine triangles, and they are precisely the corners lying on the boundary of their union — the other thirteen are interior, so no realisation can place more than fourteen corners on any circle. They are also the outer vertices of the *caturdaśāra*, the outermost of the classical enclosures, which is the ring of fourteen triangles surrounding the other eight, ten and ten; we call them the *outer corners* below. Placing all fourteen on one circle strengthens Chiodo's condition (i), which asks only that the six corners of t_3 and t_7 be concyclic: those six are among the fourteen, and the common circle is the rim.

Figure 2 shows the nine triangles one at a time against a ghost of the whole figure. It is worth looking at for a moment: the nine are of markedly different sizes and eccentricities, and the numbering by base height puts them in an order that has nothing to do with size.

3 Circular-arc Śrī Yantras

3.1 Generalised circles

Throughout, a *generalised circle* is the zero set of

$$\gamma(x, y) = a(x^2 + y^2) + bx + cy + d, \quad (a, b, c, d) \in \mathbb{R}^4 \setminus \{0\}, \quad (1)$$

taken up to scale. For $a \neq 0$ this is the circle with centre $(-b/2a, -c/2a)$ and radius $\sqrt{b^2 + c^2 - 4ad} / 2|a|$; for $a = 0$ it is a line. Working with (1) rather than with centres and radii has two advantages: lines are not a special case that has to be handled separately, and the condition that a point lie on the curve is linear in (a, b, c, d) , so the generalised circle through three given points is the null vector of a 3×4 matrix,

$$\begin{pmatrix} |P|^2 & P_x & P_y & 1 \\ |Q|^2 & Q_x & Q_y & 1 \\ |R|^2 & R_x & R_y & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = 0, \quad (2)$$

which is a determinant computation and never requires a case distinction. A straight side is the case $a = 0$ and is reached continuously, so the straight Śrī Yantra is a limit of circular-arc ones rather than a separate object.

Definition 3.1. A *circular-arc Śrī Yantra* is a realisation of the incidence structure Γ in which each of the twenty-seven sides is an arc of a generalised circle, the whole figure is symmetric about a vertical axis, and every node required to lie on a side does lie on it.

We stress that Definition 3.1 imposes nothing beyond the incidences and the symmetry. In particular the two grown rings discussed in Section 7 are not constrained, and t_3 and t_7 are not required to be inscribed in a common circle — though in the construction of Section 4 they always are, and in the rim.

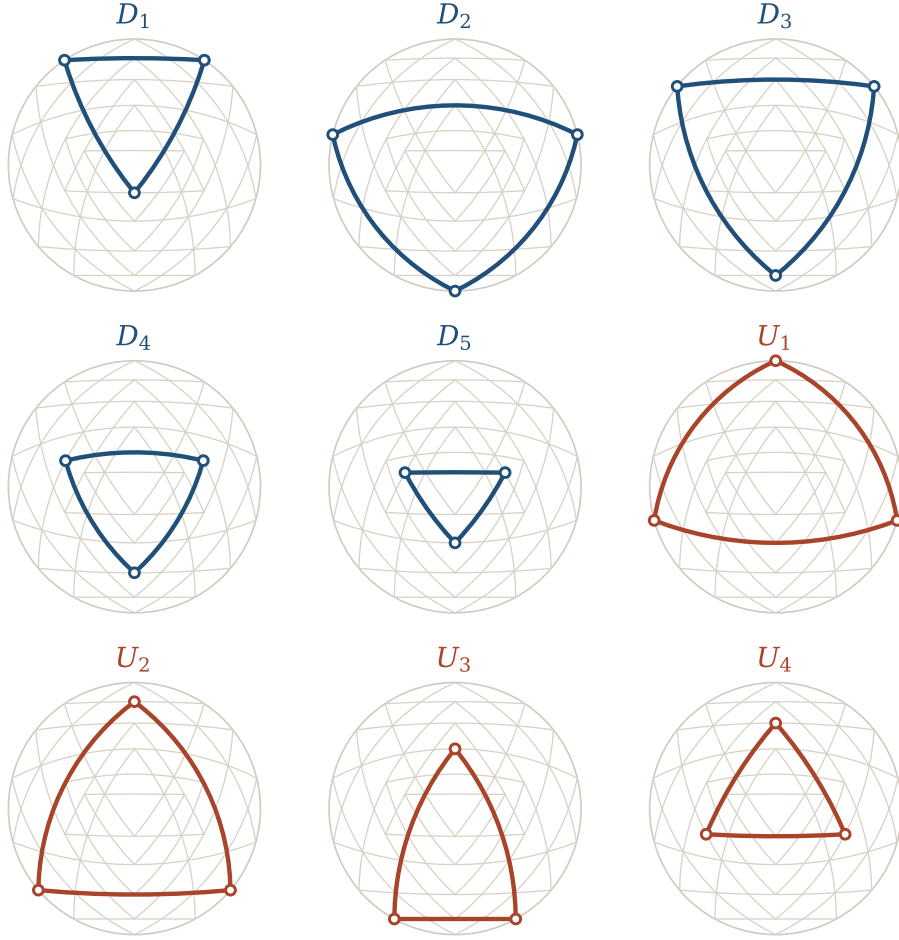


Figure 2: The nine triangles of the same figure, one at a time, in the labelling of the accompanying software (D for downward, U for upward). In Chiodo's numbering by base height these are, in order, $t_1 = D_1$, $t_2 = D_3$, $t_3 = D_2$, $t_4 = D_4$, $t_5 = D_5$, $t_6 = U_4$, $t_7 = U_1$, $t_8 = U_2$, $t_9 = U_3$, and the same correspondence holds at every member of the arc family considered here. It is not universal: the numbering follows the base heights, which move with the parameters, and on the straight figure of Section 6 the first three are instead $t_1 = D_3$, $t_2 = D_2$, $t_3 = D_1$. Open circles mark the three corners.

4 The construction

4.1 The seeds

The construction begins with twelve points, listed in Table 1 and shown on the disc in Figure 3. Eight of them are free in the plane and contribute two numbers each; one is constrained to the rim and contributes its angle; one is constrained to the axis and contributes its height; two are pinned at the poles $(0, \pm 1)$ and contribute nothing. The total is 18 free real numbers. Each seed off the axis is accompanied by its mirror image, which costs nothing further.

seed	constrained to	numbers	value in the free Default
P_{10}	free in the plane	2	$(-0.273920, 0.844094)$
P_5	free in the plane	2	$(-0.309849, -0.041519)$
P_{12}	on the rim	1	$\theta = 0.982535$
P_8	free in the plane	2	$(0.236255, 0.114118)$
P_{46}	on the axis	1	$y = 0.473624$
P_{32}	free in the plane	2	$(0.743697, 0.340431)$
P_{68}	pinned at a pole	0	$(0, +1)$
P_{60}	free in the plane	2	$(0.215835, -0.876275)$
P_{23}	free in the plane	2	$(-0.551486, -0.204177)$
P_2	free in the plane	2	$(0.190141, -0.218411)$
P_{62}	free in the plane	2	$(-0.467600, -0.021710)$
P_{55}	pinned at a pole	0	$(0, -1)$
<i>total</i>		18	

Table 1: The twelve seed points and the eighteen numbers they carry. The values are those of the distinguished member described in Section 8.

4.2 The five moves

From the seeds the figure is completed in forty-five steps, each of one of five kinds (Table 2 and Figure 4). Two produce generalised circles and three produce points:

- (B) Given a node P off the axis and a further node Q , take the generalised circle through P , its mirror image $\bar{P}$ and Q . Such a circle is automatically symmetric about the axis, and this is how the nine bases are produced.
- (C) Given three nodes, take the generalised circle through them, by (2).
- (X) Given two generalised circles, take one of their two points of intersection.
- (A) Given a generalised circle, take one of its two meetings with the axis.
- (R) Given a generalised circle, take one of its two meetings with the rim.

move	what it produces	times used
(B)	the generalised circle through a mirror pair $P, \bar{P}$ and one further point	9
(C)	the generalised circle through three points	9
(X)	the meeting of two generalised circles	16
(A)	the meeting of a generalised circle with the axis	6
(R)	the meeting of a generalised circle with the rim	5
<i>total steps</i>		45

Table 2: The five moves and how often each is used.

Moves (X), (A) and (R) each have two candidate outcomes, and the construction must say which. This is the one place where the recipe carries data beyond the eighteen parameters: each such step records an approximate location, and the branch taken is the one nearer to it. The choice is discrete, so it is locally constant in the parameters; the construction is therefore a well-defined analytic map on each connected component of the set of parameters for which no two candidate branches collide. The locus on which two branches coincide is precisely where the construction degenerates.

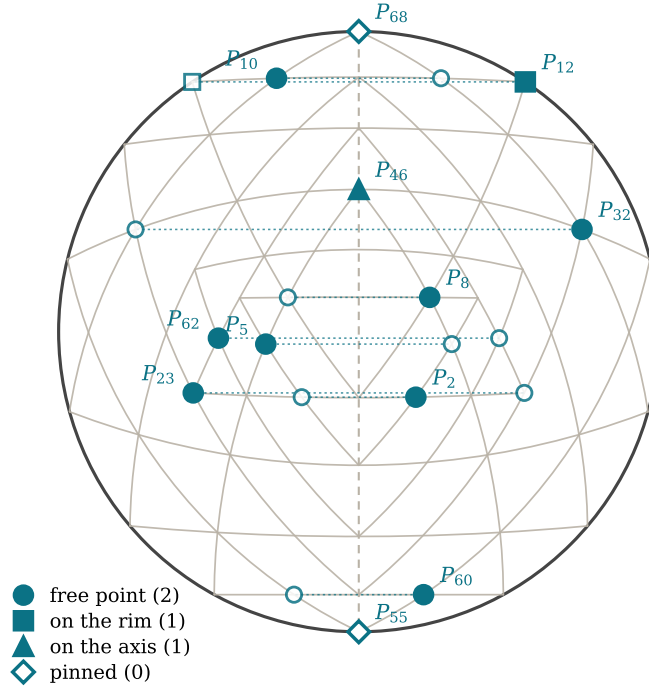


Figure 3: The twelve seeds on the disc, with the finished figure behind them in grey. Filled markers are the seeds proper; open markers are their mirror images, joined to them by dotted lines. The two diamonds at the poles are pinned and carry no parameter.

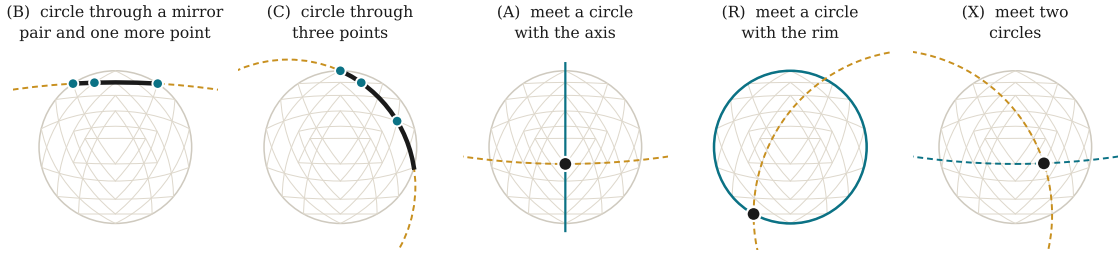


Figure 4: The five moves, each shown at the step of the construction where it first occurs. Dashed curves are the generalised circles being used, the heavy black arc or dot is what the step produces, and the figure is behind in grey.

Remark 4.1. Every move is a straightedge and compass operation. (B) and (C) are the circumscribed circle of three points, which is the classical construction by perpendicular bisectors; (X), (A) and (R) are intersections of a circle with a circle or a line. Nothing here needs the Apollonius step that Chiodo [2] requires for the straight figure, and nothing needs a conic. This is not because the arc problem is intrinsically simpler, but because the additional curvature parameters afford sufficient freedom to place the objects directly, so that no auxiliary tangency problem arises.

4.3 The order of construction

Figure 5 shows the twenty-seven sides appearing in the order the recipe produces them. The bases of the largest triangles come first, because they can be drawn from the seeds alone; the smaller triangles come last, because their corners are intersections of arcs that do not exist until the larger ones are in place.

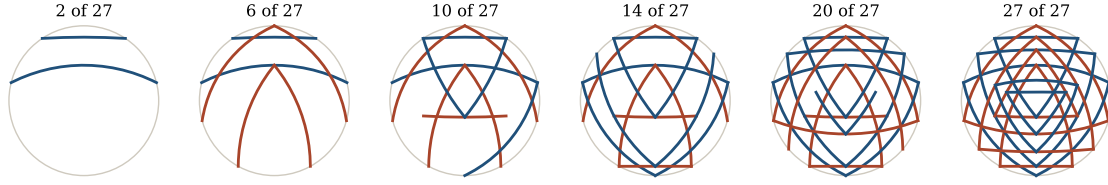


Figure 5: The construction unfolding. Each panel shows the arcs available after a further block of steps; downward triangles in blue, upward in red.

5 Exactness

The point of a construction, as opposed to a numerical fit, is that the conditions hold because of how the figure is made rather than because a residual has been driven small. For the circular-arc yantra this is true in a strong form: Chiodo’s conditions hold not merely to machine precision but *identically*, in the sense that the two objects a condition asserts to be equal are literally the same node of the configuration.

Theorem 5.1. *Let F be any figure in the image of the construction Φ of Section 4. Then F satisfies conditions (i), (ii) and (iii) of Section 2. Moreover:*

- (a) *each of the seven conditions of type (ii) holds because $\text{apex}(t')$ and $\text{bp}(t'')$ are the same node of Γ ;*
- (b) *each of the twelve conditions of type (iii) holds because a single node of Γ lies on all three of the arcs concerned;*
- (c) *condition (i) holds with the common circle equal to the rim: the six corners of t_3 and t_7 are among the fourteen nodes constrained to the unit circle;*
- (d) *the fourteen outer corners — the vertices of the caturdaśāra — all lie on the rim, and since the remaining thirteen corners are interior, no realisation of Γ places more corners on a circle.*

Proof. All four statements are about the incidence structure Γ , not about the particular parameters, so it suffices to exhibit the relevant nodes; they are recorded in Table 3. For (a), in each of the seven pairs the node listed is by definition the apex of t' — the node where the two legs of t' meet — and is also the middle node of the base chain of t'' , which is the node at which that base crosses the axis. Since the base is symmetric about the axis, its crossing with the axis is its base point, so $\text{apex}(t') = \text{bp}(t'')$ as required. For (b), in each of the twelve triples the node listed belongs to the node chain of a leg of t' , of a leg of t''' and of the base of t'' ; since Φ places every node of a chain on the corresponding arc, the three arcs share that node. Uniqueness of the common point is the statement that no two of the three arcs coincide, which holds on the open set where the construction is defined. For (c), the corners in question are the two ends of the base of t_3 , its apex, and the same three for t_7 ; the construction places all six on the rim, two of them being the pinned poles. For (d), the nodes the construction sends to the rim are the rim seed with its mirror, the two pinned poles, and the five nodes produced by move (R) with their mirrors: fourteen in all, and each of them a corner. The remaining thirteen corners lie in the interior of the union of the nine triangles, hence strictly inside any circle containing the figure. $\square$

Figure 6 shows where in the figure the incidences fall.

condition	node	condition	node
$((ii); t_8, t_1)$	67	$((iii); t_1, t_2, t_7)$	13
$((ii); t_6, t_2)$	61	$((iii); t_2, t_3, t_7)$	32
$((ii); t_9, t_3)$	46	$((iii); t_1, t_3, t_8)$	14
$((ii); t_1, t_6)$	16	$((iii); t_1, t_4, t_6)$	15
$((ii); t_5, t_7)$	3	$((iii); t_1, t_5, t_9)$	8
$((ii); t_4, t_8)$	27	$((iii); t_4, t_6, t_9)$	21
$((ii); t_2, t_9)$	36	$((iii); t_2, t_7, t_9)$	34
		$((iii); t_3, t_7, t_8)$	48
		$((iii); t_3, t_8, t_9)$	59
		$((iii); t_4, t_4, t_8)$	24
		$((iii); t_5, t_5, t_6)$	0
		$((iii); t_2, t_6, t_6)$	20

Table 3: The node of Γ that makes each of Chiodo’s conditions hold. The numbering of nodes is that of the accompanying files. Because these are statements about Γ , the same table serves every member of every family considered here.

Remark 5.2. Theorem 5.1 does not assert that the circular-arc problem is trivial. Rather, the difficulty is relocated: the conditions are free once the incidence structure is realised at all, and the whole content of the problem is in realising it — that is, in the forty-five steps, which have to be arranged so that every node lands on every arc it is supposed to. The straight case is hard because that arrangement is over-determined; the arc case is tractable because it is not.

6 The dimension of the family

Theorem 6.1. *The family of circular-arc Śrī Yantras produced by Φ has dimension 18.*

The number can be obtained in two ways which share no step, and they agree.

First count: the construction

Table 1 lists the parameters: eight free points at two numbers each, one angle on the rim, one height on the axis, and two pinned poles, giving $16 + 1 + 1 + 0 = 18$. Every later step is determined by what precedes it, so Φ is a map from an open subset of $\mathbb{R}^{18}$ to configurations. It is injective on the set where it is defined, since the seeds are recovered from the figure as particular nodes.

Second count: the incidence system

Alternatively, forget the construction and impose the incidences directly. Parametrise a symmetric configuration by the free coordinates of its nodes and the coefficients of its arcs, modulo the symmetry. Counting as in Section 2: seven nodes on the axis contribute one number each, six mirror pairs on the rim contribute one angle each, and the remaining twenty-four mirror pairs contribute two numbers each, giving $7 + 6 + 48 = 61$ node parameters. Nine of the sides are symmetric bases, each of which is a generalised circle with a horizontal tangent direction imposed and so carries two parameters; the other eighteen fall into nine mirror pairs, one representative of each carrying the three parameters of a generalised circle up to scale, giving $18 + 27 = 45$ arc parameters. The total is 106.

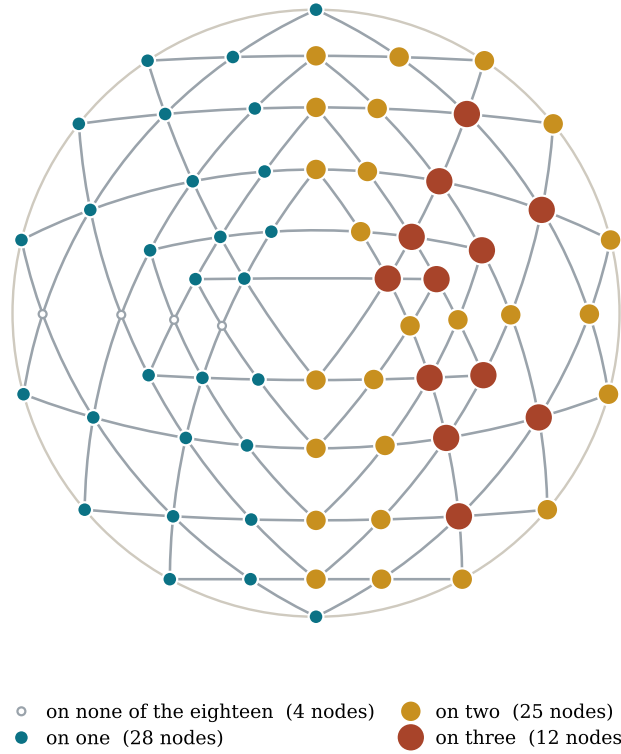


Figure 6: Where the incidences sit. Each node is drawn at a size proportional to the number of the eighteen mirror-representative sides it is required to lie on. Exactly twelve nodes carry three incidences each, and they are exactly the twelve nodes of the right-hand column of Table 3: the points at which Chiodo's condition (iii) is realised. Their mirror images carry no incidences here, because only one side of each mirror pair is represented in the system.

The incidences to be imposed are, for each of the eighteen mirror-representative sides, that each node of its chain lie on it. That is 114 scalar equations. Their Jacobian at the distinguished figure of Section 8 has rank 88, so 26 of the 114 are dependent, and the solution set is smooth of dimension $106 - 88 = 18$ there.

	count	
node parameters	61	7 axis nodes, 6 rim pairs, 24 free pairs
arc parameters	45	9 symmetric bases, 9 leg pairs
<i>unknowns</i>	106	
incidence equations	114	on the 18 mirror-representative sides
rank of the system	88	computed at the Default figure
<i>dimension</i>	18	$= 106 - 88$
free numbers in the construction	18	counted from Table 1

Table 4: The two counts. The last line is the count from the construction; the line above it is the count from the incidence system. They agree.

Remark 6.2. The agreement is a check on both. The construction could have been redundant, producing a family of dimension less than 18 because two of its parameters moved the figure the same way; the incidence system could have been miscounted, or evaluated at a non-generic point. Neither is the case. We computed the rank at several

members of the family and obtained 88 each time; we have not proved that the rank is 88 at every point, and there is no reason to expect it to be — the rank can only drop on the locus where the figure degenerates, and it is exactly there that the construction fails as well. The six members tested were the distinguished figure of each of the three families and a perturbation of each.

7 The grown rings, and two subfamilies

Outside the nine triangles the arcs continue to cross one another, and the crossings bound two further rings of cells, of fourteen cells each. We call their outer vertices *tips*.

These rings should not be confused with the classical enclosures. The traditional description — the *Rudrayāmala Tantra*, as reported by Gupta [5] and by Chiodo [2, p. 379] — speaks of the bindu, a central triangle, and enclosures of 8, 10, 10 and 14 triangles, the outermost being the *caturdaśāra* of Section 2. Those lie *inside* the nine triangles, as parts of the subdivision the triangles cut out of one another. The rings considered here are what the twenty-seven arcs bound when they are continued *outside* the nine.

Remark 7.1. That region is not empty in the traditional figure either. Between the yantra’s circle and the square *bhūpura* the painted and engraved Śrī Yantras carry two rings of lotus petals: the *aṣṭadala padma* of eight petals within, and the *ṣoḍaśadala padma* of sixteen without. These are two of the nine *āvaraṇas* into which the figure is traditionally divided, and in the received drawings they are ornament: their number and their placing are given, but nothing in the nine triangles produces them.

The grown rings occupy exactly that annulus, and they are produced by the figure — they are what the twenty-seven arcs cut out of the plane immediately beyond the triangles. Figure 7 sets the two side by side. We do not suggest a historical derivation, and the counts do not agree: the grown rings have fourteen cells each, against eight and sixteen petals. But a ring of pointed cells standing on a circle and reaching to another is the shape a lotus ring has, and the construction supplies one without being asked. The classical requirement that each *āvaraṇa* close on a circle is, in this reading, the same requirement as concyclicity of a ring’s tips, which is the condition of Theorem 7.2.

For completeness we record the interior subdivision. The nine triangles cut the disc into 74 atomic regions: the central one, and then shells of 3, 8, 8, 10, 10, 10, 10 and 14 reading outwards. This is finer than the classical count of 43, which groups some of these regions together; the decomposition is the same for the straight figure and for every member of the arc family, since it depends only on Γ . In a general member of the family the tips of a ring are not concyclic, and they are not nearly concyclic either: at the distinguished figure of the free family the radii of the first ring’s tips span 22.3% of their mean and those of the second ring 135.8%. Concyclicity is a real restriction, not a small correction.

Asking for them to be concyclic is a natural condition, and it is the arc analogue of the traditional requirement that the enclosures close on circles.

Theorem 7.2. *Within the 18-dimensional family:*

- (a) *the condition that the fourteen tips of the first ring be concyclic has rank 6, and cuts out a subfamily of dimension 12;*
- (b) *the condition that the fourteen tips of the second ring be concyclic has rank 7, and cuts out a subfamily of dimension 11.*

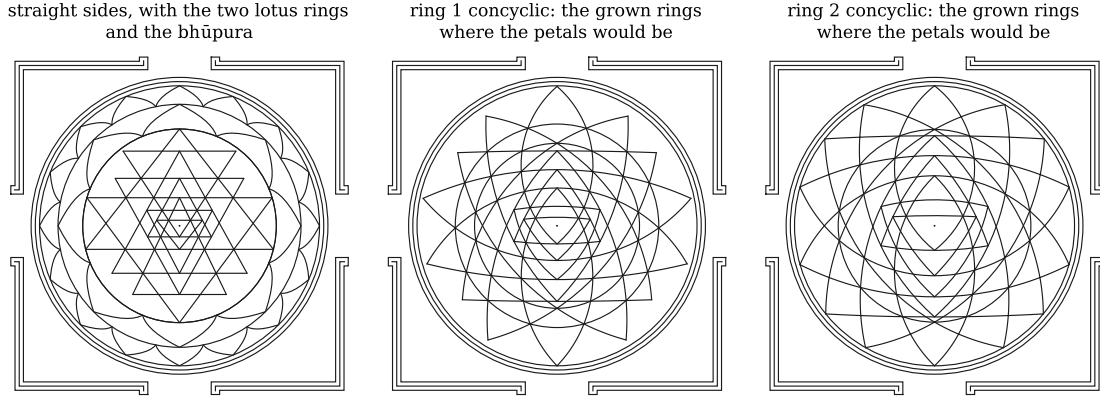


Figure 7: Left: the figure as traditionally drawn, with straight sides and the eight- and sixteen-petalled lotus rings. Middle and right: the two ring-held members of the circular-arc family, without petals, the grown rings occupying the same annulus. In the middle panel the circle carrying the first ring’s tips is drawn explicitly, that ring lying interior to the second; in the right-hand panel the second ring’s tips are of maximal radius, so the innermost frame circle already passes through them. Each panel carries the same frame, in an open form of the *bhūpura*: three concentric circles and a band broken at the middle of each side, its four corners closing on themselves. The innermost circle passes through the outermost points of each figure — the outer petal tips on the left, the outer ring tips on the other two.

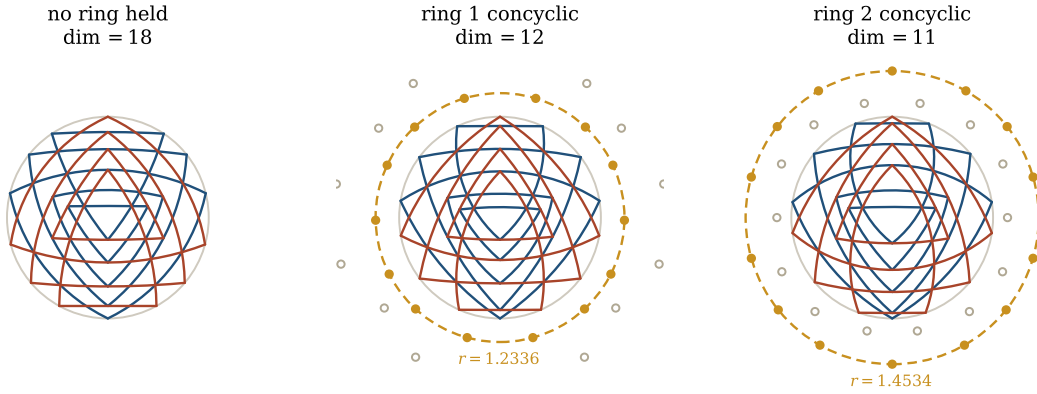


Figure 8: The three families. Left: no ring held, the general case, dimension 18. Middle and right: the first and the second grown ring required to be concyclic, dimensions 12 and 11. Filled dots are the tips of the held ring and the dashed circle is the circle they lie on; open dots are the tips of the other ring, which are not concyclic.

Figure 8 shows the three families side by side.

The ranks are computed from the Jacobian, in the eighteen seed parameters, of the fourteen residuals “radius of tip i minus the mean radius”. That these ranks are 6 and 7 rather than 13 is mirror symmetry doing most of the work: the fourteen tips fall into mirror classes, and within a class the radii are equal automatically.

Remark 7.3. The ranks quoted are computed *on* the locus in question, and this matters. Evaluated instead at a general member of the 18-dimensional family, where neither ring is concyclic, the first ring’s system has rank 7 and the second’s rank 8. The ranks therefore drop by one on each of the two loci: these subfamilies are precisely where the conditions become degenerate. The dimensions 12 and 11 quoted in Theorem 7.2 are the ones that match the construction, in the sense that the tangent space computed from the seeds at a

ring-held figure has exactly that dimension.

Remark 7.4. The two conditions are not nested: holding the first ring does not hold the second, nor conversely. Whether a circular-arc Śrī Yantra can satisfy both simultaneously is not addressed here. We note only that a rank computed at a figure satisfying neither condition would not settle the question, for the reason given in Remark 7.3: the ranks of these systems fall on the loci where the conditions hold.

7.1 Internal constructions and arranged coincidences

Theorem 7.2 locates the two subfamilies but says nothing about how a member of one might be drawn. This subsection and the next take up that question. The conclusion is two-sided: no construction of the kind used in Section 4 places a ring on a circle, and that obstruction can be exhibited in a form the reader can check by hand; but a construction that departs from it at exactly one point does exist, for each ring, and the departure is of the same kind as the one Chiodo’s construction of the straight figure makes.

We fix the vocabulary. A *construction* is a sequence in which each circle, each node and each ring tip of the figure is produced exactly once, the rim, the axis and the ring’s circle being given. Each object has a *need*: the number of prior incidences that determine it. A general circle needs three of its points and an axis-symmetric base needs two; a node needs two of its circles, or one if it is confined to the axis or the rim, or none if it is a pole; a ring tip needs one of its two circles, or none if it is the point where the ring’s circle crosses the axis. An object may be produced when exactly *need* of its incidences are already available. A node may instead be placed as a *seed*, from no circle at all, at a cost of two parameters in the plane and one on the axis or the rim. A construction is *internal* if it draws nothing that is not a part of the figure; the construction of Section 4 is internal, and so is everything considered below.

The cost of a construction is not a matter of choice. Every incidence between a node and a circle is satisfied at whichever of the two is produced later — the circle is drawn through the node, or the node is cut from the circle — so the number of parameters spent is the sum of the needs less the number of incidences, the same for every order. What varies from order to order is only whether some object is reached with *more* than its need already present. Such an object has one incidence too many: it is drawn through as many points as determine it, and a further point of the figure must then happen to lie on it. We call this an *arranged coincidence*. It costs one parameter, which is spent on making the coincidence hold, and it raises by one the seed weight the construction can afford, so that

$$\text{seed weight} = \dim + \#\{\text{arranged coincidences}\}, \quad \text{free parameters} = \dim. \quad (3)$$

A construction with arranged coincidences still sweeps the whole family; only the route into it changes. Zero arranged coincidences is the internal case.

Proposition 7.5. *No internal construction places either ring on a circle. The same is true of the straight figure.*

Proof. Whatever object is produced last has every one of its incidences already present, so it must have at most *need* of them in all. Remove it and repeat. If the process stops before the figure is exhausted, what remains is a set $\mathcal{K}$ of objects each of which meets more than its need *within* $\mathcal{K}$, and then whichever of them is produced last — in any order whatever — has one incidence too many. The set $\mathcal{K}$ does not depend on the removal order, since an object removable at one stage remains removable at every later one.

For the first ring $\mathcal{K}$ has 36 objects: 17 nodes, 12 circles and all 7 tips. For the second it has 37, the eighth tip included. For the straight figure it has 32: 18 nodes and 14 sides. The lists are produced by `core.py` in the repository of Section 8; each takes a fraction of a second, and each may be checked against the labelled figures by counting incidences. $\square$

The proof deserves two comments. First, it is not a shortage of freedom. Inside the 36 objects of the first ring's core there are 52 incidences against needs summing to 58; for the second ring 52 against 57; for the straight figure 44 against 48. Every subset of every one of the three figures has at least as much need as incidence — we verified this by a maximum-flow computation — so a counting argument alone would allow an internal construction. What forbids it is circular: within $\mathcal{K}$ every object is over-determined by the others, as the three edges of a triangle over-determine each vertex although three edges is not too many for three vertices. Second, we had earlier established the proposition for the two rings by exhaustive search, with a conflict-driven clause-learning solver (`z3` [10]) splitting the order into 84,211 disjoint parts and refuting each. The certificate above reproduces that result, and the straight case as well, without a solver.

Proposition 7.6. *For each ring, and for the straight figure, a construction with exactly one arranged coincidence exists. Each has been carried out numerically and every incidence of the resulting figure verified to 10^{-15} ; the identity (3) holds in each case, with seed weights 12, 11 and 5 and free parameters 11, 10 and 4.*

The orders were found by the same solver, now permitted one over-determined object. Since Proposition 7.5 forbids zero, one is the minimum. The straight case is the calibration: Chiodo's construction contains exactly one step that is not forced by what precedes it — the circle–line–point problem of Apollonius by which he fixes a height — and the count of Proposition 7.6 agrees with it. In this bookkeeping the two ring-held arc figures and the classical straight one are of the same character: each is an internal construction that fails at one point, and at that point a single number must be chosen so that a single coincidence holds.

The viewer of Section 8 draws the two ring-held families by such a construction: eleven parameters (ten for the second ring) and the ring's radius are free, and the one coincidence is solved by bisection at run time, every other step being a circle through points or a meeting of two circles. Among the several hundred orders the solver produces, all equal as mathematics, we kept in each case the one under which the figure moves least for a unit change of its parameters, since this decides how far a parameter can be moved before some step of the construction ceases to be defined; the choice widens that range by two orders of magnitude.

7.2 Whether the coincidence can be drawn

What Proposition 7.6 leaves open is whether the one arranged coincidence can be arranged by ruler and compass, as Chiodo's is, rather than by solving. We state the question precisely and report what we have been able to establish about it. The answer is not known to us.

Fix an order with one arranged coincidence, and let t be the parameter that is solved for. Everything that does not depend on t is already drawn, from freely chosen seeds, before the coincidence is reached; call the field these objects generate K . The coincidence is one equation in t over K , and its root t^* can be produced by ruler and compass if and only if t^* lies in the quadratic closure of K . This formulation already settles one question. A construction that draws auxiliary objects not belonging to the

figure — Chiodo’s Apollonius circle is such an object — produces nothing outside that closure, so the restriction to internal constructions in Proposition 7.5 costs nothing here: whether the ring-held figures are constructible with straightedge and compass at all is exactly the question of whether t^* is constructible over K , and no widening of the class of constructions changes it.

Three observations bear on that question.

Lines and circles. Chiodo’s method does not transplant, and the reason is elementary. In the straight figure the sides are lines, and 28 of the 39 node orbits are the meeting of two lines, a rational operation; only the four outer corners cost a square root. From one of his heights to the coincidence he must arrange, the chain meets no square root before the circle itself, and the condition it arrives at is quadratic: it is the circle–line–point problem of Apollonius, which is why an auxiliary construction can solve it. In the arc figure every node not pinned by the frame is the meeting of a circle with a circle, the rim or the axis — a square root each — and the ring tips add seven more. There is no rational path from any parameter to any coincidence. Quantitatively, define the *depth* of a parameter as the number of square roots that both depend on it and enter the coincidence. For the two rings, over two hundred orders each, its minimum is 7, and for the second ring the same seven objects are responsible in every order that attains it. For the first ring’s chosen order we computed the product of the coincidence over all 2^7 branches of those roots, which is a rational function of t whose numerator is the polynomial t^* satisfies over K : it has degree greater than 128. Neither fact excludes constructibility — a root of a polynomial of high degree may still lie in a quadratic tower — but both put the direct algebraic route beyond reach.

Where the coincidence can sit. The core $\mathcal{K}$ of Proposition 7.5 also says where the one over-determined object may be. Allowing a single object one incidence more than its need, and repeating the removal, the core is cleared by exactly 11 of its 36 members for the first ring and 11 of its 37 for the second: three circles, two interior crossings, three nodes confined to the rim or the axis, and three ring tips. Pinning the coincidence to each of these in turn and asking the solver for orders, every one of the eleven is realised, by several hundred orders each. The interesting ones are the nodes on a locus and the ring tips. A coincidence there reads: *two circles and this locus are concurrent*. If one of the two circles does not depend on t , it cuts the locus in a constructible point X ; the coincidence then says only that the other circle passes through X , and if that circle is drawn through points of which all but one are independent of t , it is the circle C^* through those points and X , constructible in turn. The remaining point Q is then the meeting of C^* with whatever curve Q runs along as t varies — constructible if that curve is the axis, the rim, the ring’s circle, or a circle independent of t . This is the shape of Chiodo’s step, transposed to arcs.

We tested every order collected at every one of the eleven sites, for every parameter the coincidence responds to — more than fifty thousand pairs — determining what does and does not depend on t by displacing t and recomputing, rather than by inference. The shape occurs. At one node of the second ring it occurs 149 times, and at one tip of the first ring 14 times, always in the same configuration: a still circle cuts the locus, the axis-symmetric circle through the resulting point and one further still point is C^* , and Q is the meeting of C^* with a still circle. Three compass strokes to a constructible point. In every instance, however, the point so reached is not the parameter. The parameter is a seed two steps further back, behind a circle drawn through three points of which two

depend on it, and nothing in the figure provides a second curve along which to cut it. The chain reaches a point and stops. At the other nine sites it does not begin: at the circles, the coincidence circle itself depends on t in every order; at the interior crossings and at the axis tip, the objects that would make the point constructible never all stand still.

Special members. The considerations above concern a generic choice of the free parameters. Since a single drawable figure would suffice, we also searched for one: taking the free parameters and the radius rational — each trial rationalises them one at a time with the smallest denominator that leaves a figure standing, so that every trial is a genuinely rational point of the family — and testing whether the solved parameter is rational, quadratic, or a quartic with a rational resolvent, to forty digits. Two hundred thousand trials per ring found none. We record this for completeness; a search of this kind cannot succeed except by luck and its failure is not evidence.

Remark 7.7. What is proved is therefore this. Both ring-held families are reached by an internal construction with one arranged coincidence, and not by fewer. The coincidence may be placed at any of eleven objects and at no other. At two of those it is, in a definite sense, three compass strokes from being drawn, and in every order examined the last of the three does not close. Whether some order closes it, or whether the number it determines lies outside the quadratic closure of the drawn figure, is the question we leave open. It is a question about a field, not about an order of drawing; and in that respect the ring-held Śrī Yantras stand where the straight one stood before Chiodo, with the difference that the elementary route he found is, for the reason given under *Lines and circles*, not available here.

8 The accompanying software and GeoGebra files

An interactive viewer accompanies this paper, together with the three GeoGebra files described below. Both are available at

<https://github.com/michaelkopp/sriyantra>

The viewer draws any member of the three families from its eighteen parameters, in the principal directions of Section 6 or in the seed coordinates themselves, and exports the figure as SVG or PNG. The viewer also draws Chiodo’s straight side construction. The yantras displayed in Figure 7 can be reproduced with the viewer by exporting the default presets.

The three GeoGebra files are:

file	family	dimension
sri-yantra-arcs-free.ggb	no ring held	18
sri-yantra-arcs-ring1.ggb	first ring concyclic	12
sri-yantra-arcs-ring2.ggb	second ring concyclic	11

Each carries one distinguished member of its family, the one used for the figures above and for the numerical statements in the text. The files are not static drawings. Each contains a construction, written out as GeoGebra [4] commands and evaluated when the file is opened, the seeds as free or constrained points and every later step a **Circle** through points or an **Intersect** of two objects, followed by the twenty-seven sides as circular arcs. The first file carries the construction of Section 4: twelve seeds and forty-five steps.

The two ring files carry the constructions of Proposition 7.6: eleven free numbers, or ten, together with the radius R of the ring's circle, and some sixty steps in which the tips of the held ring are cut from that circle. All three carry the grown rings, each ring cell drawn from the same twenty-seven circles continued outside the yantra, its tip where two of them cross; ring 1's cells rest on nodes of the yantra and ring 2's on the tips of ring 1. The ring files also carry the circle the held ring sits on, drawn in red.

Dragging a seed re-derives the whole figure, rings included, and the reader who does so will see the difference between the two kinds of file. In the first, the incidences of Theorem 5.1 survive — they are steps of the construction — and nothing else is asked. In the ring files the concyclicity survives as well, since the tips are produced on the circle; what does not survive is the one arranged coincidence. Each ring file names it in its opening comment and carries a number `gap`, the distance of the point concerned from the circle it is required to lie on: zero at the values written into the file, and not zero after a drag. That number is the whole difference between what Proposition 7.6 provides and a construction with straightedge and compass.

The branch choices of moves (X), (A) and (R) are carried in the files as the nearest-point rule described in Section 4, so that a small drag keeps the intended branch. A large drag can cross a wall where two branches meet, after which the figure will jump; this is a real feature of the construction rather than an artefact of the file.

We verified the files by re-evaluating their command lists in an independent implementation of the same operations — circles through three points by (2), conic intersections by the radical line — and comparing all sixty-nine nodes with the values produced by the construction. The largest disagreement is 2.2×10^{-12} for the first file and 2.1×10^{-10} for the ring files, consistent with the twelve decimal places to which the seed values are written and with the longer chains of the ring constructions, which amplify that rounding. In the ring files the fourteen tips of the held ring lie on its circle to 2×10^{-16} , as they must, being cut from it.

The computations of Sections 7.1 and 7.2 are carried out by scripts in the same repository, each standing alone and documented in its header. `core.py` produces the certificate of Proposition 7.5 for all three figures in under a second and needs nothing beyond Python. `fits.py` finds constructions with a prescribed number of arranged coincidences, pinned if desired to a given object, and replays any saved order against the rules; it needs `z3` for the search but not for the replay. `ring1_cubes.py`, `ring2_cubes.py` and `chiodo_cubes.py` are the earlier solver refutations. `concurrent.py` is the test of Section 7.2, and `hunt.py` the search for special members.

Software

The construction, the figures and the numerical statements in this paper were produced with the viewer at <https://github.com/michaelkopp/sriyantra>, which is distributed with the three GeoGebra files.

9 Concluding remarks

The circular-arc Śrī Yantra is a considerably less rigid object than the straight one. Where Chiodo must solve an Apollonius problem to close the straight figure, the free arc figure closes by elementary steps, and the concurrency conditions that make the straight problem delicate are here consequences of the incidence structure alone. Holding a ring brings the

difficulty back, in the same form and the same amount: one coincidence, at one point. The interest moves from whether the figure can be drawn to which of the eighteen-parameter family one should choose, and it is here that the classical conditions — the enclosures closing on circles — return as genuine constraints, cutting the family to twelve and to eleven.

Four questions remain.

1. Each ring-held figure is drawn by a construction that is ruler and compass except at one point, where a number is chosen so that a coincidence holds. Can that number be produced by ruler and compass? Section 7.2 reduces this to whether one algebraic number lies in the quadratic closure of the field the drawn figure generates, and shows the coincidence can sit at only eleven objects. The straight figure's analogous number is a root of a quadratic, which is Chiodo's Apollonius problem; here every path from a parameter to the coincidence passes through at least seven square roots, and the polynomial the number satisfies has degree above 128. A negative answer would be a theorem of some interest, since it would separate the ring-held arc figures from the straight one in kind and not only in degree.
2. The straight Śrī Yantra is the locus $a = 0$ for all twenty-seven arcs, which is 27 conditions on 18 parameters and so is not reached inside this family. The straight and arc constructions are therefore not specialisations of one another, although both realise Γ . Is there a construction containing both? We conjecture that conic sides suffice: a conic through five points carries two parameters more than a circle through three, and the straight figure would arise as a degeneration within one connected family. The incidence structure would require re-examination, a conic through five points being a step of a different character.
3. The fourteen outer corners, the vertices of the *caturdaśāra*, are the only ones that can meet a circle containing the figure; which fourteen they are depends only on Γ . Call a realisation *geodesic* if its twenty-seven sides are geodesics of a Riemannian surface, and let $\nu \leq 14$ count those outer corners lying on the circumscribed circle. The construction of Section 4 attains the maximum $\nu = 14$, with no corner outside, but is not geodesic. In the Euclidean plane the geodesic realisations are the straight figures, where ν depends on the member: $\nu = 6$ for that of Section 6, the corners of t_3 and t_7 supplied by condition (i), and $\nu = 10$ for Kulaichev's [7], which puts the ends of the bases of t_1 and t_9 on the circle as well. Rao [9] treats the spherical case but optimises against other criteria, so the maximum of ν there is not settled by his figures. Is there a Riemannian surface admitting a geodesic realisation with $\nu = 14$, and is any such realisation aesthetically satisfying?
4. Which member of the family should be selected? This is not, in the end, a mathematical question. The criterion governing the form of the Śrī Yantra has always been in part aesthetic, and while clarity and uniqueness of construction contribute to the quality of a figure, the two need not agree: optimisation of a metric quantity in isolation readily yields figures of little contemplative value. The distinguished members displayed in this article and carried by the accompanying files were selected on aesthetic grounds.

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